

Charged lepton mass matrix with democratic family mixing

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Received 24 April 1988; in revised form 25 May 1989

Abstract. By starting from a mass matrix with family-independent mixing (the so-called “democratic” family mixing) and by requiring some simple constraints on the additional mass terms, but without any small parameters, a mass relation which is in excellent agreement with experiment is derived for charged leptons.

What is of great interest to us is how to understand the observed hierarchical pattern of the quark and lepton mass spectra: why the third generation quarks and leptons are extremely heavy compared with the other two generations; why the first generation masses are negligibly small.

Recently, Fritzsch [1] has proposed an attractive idea that quark and lepton mass matrices are given by a mass matrix M_D in which all matrix elements are equal

$$M_D = \frac{1}{3} m \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad (1)$$

and some additional small mass terms.

Jarlskog [2] has named the mass matrix of the type (1) the “democratic” family mixing. As well-known, the mass matrix (1) can be diagonalized by the transformation matrix A

$$A = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix} \quad (2)$$

as

$$AM_D A^\dagger = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & m \end{pmatrix}. \quad (3)$$

The mass matrix with the form (1) has originally introduced [3] in order to describe the neutral meson ($\pi^0 - \eta - \eta'$) masses and mixing for any J^P .

Mass matrix models with the form (1) for quarks have already proposed by Harari et al. [4] and the author [5, 6]. For recent studies of the democratic family mixing models, see [7, 8].

Now, Fritzsch asserts that lepton mass matrix, too, should be started from the mass matrix of the type (1).

Since the down-lepton (charged lepton) mass values are well established experimentally, the experimental check of the models can be decidedly done for down-lepton mass matrix. If we can get a successful model for down-lepton mass matrix, it will provide a strong clue to the investigation of the unification model for the quark and lepton mass matrices. Therefore, in the present paper, we concentrate our interest on down-lepton mass matrix only.

According to Fritzsch's idea, we start from the lepton mass matrix of the type (1).

In a step subsequent to the mass matrix with the type (1), electron (the first generation down-lepton) should still be exactly massless. As such a simple example of the democratic family mixing model, we assume the following form:

$$M_o^{1/2} = \frac{1}{3} a_0 \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} + b_0 \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (4)$$

Here, we should note that we assume the form (4) not for the linear mass matrix M , but for the square root mass matrix $M^{1/2}$, where $M^{1/2}$ is defined by $M = (M^{1/2})^2$.

At present, there is no reason that we should start

not from M but from $M^{1/2}$ with the form (4). This is simply an ansatz. For example, for the mass generation mechanism based on the extended technicolor [9], it is likely that that mass matrix M is given by the form $M = (M^{1/2})^2$. However, in this paper we do not discuss the origin of the form $M = (M^{1/2})^2$, and we start our study from the ansatz (4). The purpose of the present paper is to point out that if we start from the mass matrix $M_0^{1/2}$ with the form (4), we can obtain a beautiful relation for the charged lepton masses under some simple constraints on the coefficients a_0 and b_0 .

First, we assume that in the limit of $m_e = 0$, i.e., in the mass matrix (4), the coefficients a_0 and b_0 are determined as the sum of m_τ and m_μ , $\text{Tr}[D_0]$, takes a minimum under the condition

$$\begin{aligned} \sqrt{m_\tau} + \sqrt{m_\mu} &= \text{Tr}[D_0^{1/2}] = \text{Tr}[M_0^{1/2}] \\ &= a_0 + b_0 = \text{constant}, \end{aligned} \quad (5)$$

where $D_0^{1/2}$ is defined by $D_0^{1/2} = D^{1/2}$ ($m_e = 0$) and

$$D^{1/2} = \begin{pmatrix} \sqrt{m_e} & 0 & 0 \\ 0 & \sqrt{m_\mu} & 0 \\ 0 & 0 & \sqrt{m_\tau} \end{pmatrix}. \quad (6)$$

Then, from the relation

$$\begin{aligned} m_\tau + m_\mu &= \text{Tr}[D_0] = \text{Tr}[M_0] = a_0^2 + b_0^2 + \frac{2}{3}a_0b_0 \\ &= \frac{2}{3}(a_0 + b_0)^2 + \frac{1}{3}(a_0 - b_0)^2, \end{aligned} \quad (7)$$

we get

$$a_0 = b_0. \quad (8)$$

(The result (8) can also be derived from the assumption that $\sqrt{m_\tau} + \sqrt{m_\mu}$ and $m_\tau + m_\mu$ is stable under displacement of the coefficients $a_0 \rightarrow a_0 + a_1\epsilon$ and $b_0 \rightarrow b_0 + b_1\epsilon$.) This result (8) suggests not that the second term (b_0 -term) in (4) appears after the first term (a_0 -term), rather, but that the both terms appear simultaneously at the same energy level.

In this stage, we get the mass ratio

$$\left. \frac{m_\tau}{m_\mu} \right|_{m_e=0} = \left(\frac{\sqrt{3}+1}{\sqrt{3}-1} \right)^2 = 13.9. \quad (9)$$

Next, we consider a small deviation term from the mass matrix (4) so that electron acquires a very small mass. We assume that the violation of the democratic family mixing is caused by the form

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1+\epsilon_a & 1+\epsilon_a \\ 1+\epsilon_a & 1 & 1+\epsilon_a \\ 1+\epsilon_a & 1+\epsilon_a & 1 \end{pmatrix}. \quad (10)$$

We consider that the second term in (4) also comes from a democratic family mixing in another down-lepton expression, so that we consider a small violation term of the type (10) for the second term, too [we also define a small parameter ϵ_b similar to (10)]. Therefore,

the general mass matrix is given by the form

$$\begin{aligned} M^{1/2} &= \frac{1}{3}a \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} + \frac{1}{3}bA \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} A^\dagger \\ &+ \frac{1}{3}c \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \end{aligned} \quad (11)$$

where $a = a_0(1 + \epsilon_a)$, $b = b_0(1 + \epsilon_b)$ and $c = -(a_0\epsilon_a + b_0\epsilon_b)$. From the assumption (10), we get the relation

$$\text{Tr}[M^{1/2}] = a + b + c = a_0 + b_0 = 2a_0. \quad (12)$$

If we assume that the sum of the lepton masses remains invariant under the small violation, i.e.,

$$\begin{aligned} \text{Tr}[D] &= \text{Tr}[M] = a^2 + b^2 + \frac{1}{3}c^2 + \frac{2}{3}(ab + bc + ca) \\ &= \text{Tr}[D_0] = \text{Tr}[M_0] = \frac{8}{3}a_0^2, \end{aligned} \quad (13)$$

we can obtain a sum rule

$$\text{Tr}[D] = \frac{2}{3}(\text{Tr}[D^{1/2}])^2, \quad (14)$$

by eliminating a_0 from (12) and (13), i.e.,

$$m_\tau + m_\mu + m_e = 4(\sqrt{m_\tau m_\mu} + \sqrt{m_\mu m_e} + \sqrt{m_e m_\tau}), \quad (15)$$

which predicts tau lepton mass

$$m_\tau = 1776.97 \text{ MeV} \quad (16)$$

from the input data [10] m_e and m_μ . The prediction is in fairly good agreement with the experimental value [10]

$$m_\tau = 1784.1_{-3.6}^{+2.7} \text{ MeV}, \quad (17)$$

If, instead of the assumption (13), we put another assumption that the sum of the mass gaps from the lowest mass $m_e = (c/3)^2$, $(m_\tau - m_e) + (m_\mu - m_e)$, remains invariant, i.e.,

$$\begin{aligned} \text{Tr}[D] - 3m_e &= a^2 + b^2 + \frac{2}{3}(ab + bc + ca) = \text{Tr}[D_0] \\ &= a_0^2 + b_0^2 + \frac{2}{3}a_0b_0 = \frac{8}{3}a_0^2, \end{aligned} \quad (18)$$

we obtain another sum rule

$$(\sqrt{3}-1)\sqrt{m_\tau} - (\sqrt{3}+1)\sqrt{m_\mu} = 4\sqrt{m_e}, \quad (19)$$

from (12) and (18). The sum rule (19) was originally found semi-empirically in another model with technicolored SU(3)-family **8** + **1** bosons by the author [6]. The sum rule (19) predicts

$$m_\tau = 1786.57 \text{ MeV}, \quad (20)$$

which is in excellent agreement with the experimental value (17).

If we express the conditions (13) and (18) as

$$\begin{aligned} m_\tau + m_\mu + nm_e &= a^2 + b^2 + \frac{n+2}{9}c^2 + \frac{2}{3}(ab + bc + ca) \\ &= \frac{8}{3}a_0^2, \end{aligned} \quad (21)$$

we obtain the relations

$$a + b = 2a_0 - c, \quad (22)$$

$$ab = a_0^2 - 2ca_0 + \frac{n+5}{12}c^2, \quad (23)$$

so that a and b are expressed by the parameter c as follows:

$$\begin{pmatrix} a \\ b \end{pmatrix} = a_0 \left(1 - \frac{1}{2}\varepsilon^2 \pm \sqrt{\varepsilon^2 - \frac{n+2}{12}\varepsilon^4} \right), \quad (24)$$

where we have put $c = a_0\varepsilon^2$. Only for $n = -2$, i.e., for the case of the condition (18), the coefficients a , b and c are expressed in very simple form

$$\begin{aligned} a &= a_0(1 + \varepsilon - \frac{1}{2}\varepsilon^2), \\ b &= a_0(1 - \varepsilon - \frac{1}{2}\varepsilon^2), \\ c &= a_0\varepsilon^2. \end{aligned} \quad (25)$$

Although, at present, we have no reason that we should adopt the assumption (18) (i.e., $n = -2$) prior to the assumption (13) (i.e., $n = 1$), we take a great interest in the case of $n = -2$ because of the phenomenological success (20).

In any case, we can diagonalize the mass matrix (11) by the transformation matrix U

$$U = R(\theta)A, \quad (26)$$

$$R(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix}, \quad (27)$$

$$\tan 2\theta = \frac{2\sqrt{2}}{3a/b - 1}, \quad (28)$$

as

$$UM^{1/2}U^\dagger = D^{1/2}. \quad (29)$$

For the case of $n = -2$, we get $\varepsilon = 0.28$ and $a/b = 1.84$, so that we get $\theta = 16^\circ$.

Note that we cannot regard the value $\varepsilon = 0.28$ (and even $2c/(a+b+c) = \varepsilon^2 = 0.08$) as a negligibly small parameter any more. As a result, our model does not include any negligibly small parameters in spite of the observed small mass ratio $m_e/m_\mu = 0.004837$. This suggests that the “bias” term (c -term) (therefore, ε -terms in the a - and b -terms) in (11) appears at the same or near energy scale at which a_0 - and b_0 -terms appear.

In this paper, we have treated the down-lepton mass matrix only. If we assume the similar model for the down-quark mass matrix, for example, with $n = -2$, we obtain a similar sum rule to (19)

$$(\sqrt{3}-1)\sqrt{m_b} - (\sqrt{3}+1)\sqrt{m_s} = 4\sqrt{m_d}. \quad (30)$$

The sum rule is satisfied with the so-called current quark mass values [11] $m_b = 5.3 \pm 0.1$ GeV, $m_s =$

175 ± 55 MeV and $m_d = 8.9 \pm 2.6$ MeV (at $\mu = 1$ GeV) within the experimental and theoretical errors. (For example, the input values $m_b = 5.0$ GeV and $m_s = 200$ MeV predict $m_d = 10.77$ MeV.) However, if we assume the similar model for the up-quarks, we get an unsatisfactorily low value $m_t = 22.3$ GeV (at $\mu = 1$ GeV) of the predicted top quark mass from the conventional current quark mass values [11] $m_c = 1.35$ GeV and $m_u = 5.1$ MeV. Besides, we cannot explain the Cabibbo mixing because the Kobayashi–Maskawa mixing matrix V is given by $V = U_u U_d^\dagger = R(\theta_u)R(\theta_d)^\dagger$ and $R(\theta)$ provides only $2 \leftrightarrow 3$ family mixing. We must consider another mass matrix model apart from the form (11), at least, for the up-quarks.

In this paper, we do not discuss neutrino mass matrix, too, because it should be studied in connection with the relationship between up- and down-quarks. At present, the connection of our down-lepton mass matrix (11) with up-lepton mass matrix and up- and down-quark mass matrices is not clear.

In conclusion, we have proposed a charged lepton mass matrix with a democratic family mixing, which provides an excellent prediction of the tau lepton mass (20). We have, in this paper, reported a model for down-lepton mass matrix separately from up- and down-quark and up-lepton mass matrices. As a model for quark mass matrices based on the democratic family mixing, for example, the Harari–Haut–Weyers model [4] can be a promising candidate. Although our discussion is merely numerology and we have not given any field-theoretical bases for the mass matrix (11), the simplicity of our ansatz and the form (11) and the excellent phenomenological success (20) are reasons enough that it may be taken seriously. Possible origins of the democratic family mixing have, for example, been discussed by Fritzsch in [1].

Acknowledgements. The author express his sincere gratitude to Prof. C. Jarlskog for valuable comments and pointing out an erratum in an earlier version of this work. This work has been supported in part by a Shizuoka Prefectural Government Grant.

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